2. Requirement Analysis Phase

FitBuddy – AI Fitness Plan Generator using Gemini Models

Phase Report

Prepared from the supplied FitBuddy project documentation and organized into a project lifecycle deliverable.

1. Functional Requirements

FR-01: Display a home page with a user input form. FR-02: Accept username, user ID, age, weight, fitness goal, and workout intensity. FR-03: Generate a personalized 7-day workout plan. FR-04: Generate a nutrition/recovery tip. FR-05: Accept feedback and update an existing plan. FR-06: Store users and plans in SQLite. FR-07: Display registered users and their original and updated plans in an admin view.

2. Non-Functional Requirements

Usability: clear forms and readable results. Maintainability: modular Python files. Reliability: validate form inputs and handle API/database errors. Performance: use a fast model for short tips. Privacy: keep API keys in environment variables and avoid exposing secrets in templates or logs.

3. User Inputs

Username; user ID; age; weight in kilograms; fitness goal such as weight loss, muscle gain, flexibility, or general wellness; intensity selected from low, medium, or high; feedback text for plan revision.

4. System Outputs

A seven-day workout schedule with exercise details, sets/repetitions or duration, warm-up and cooldown guidance, a nutrition/recovery tip, and an updated plan after feedback.

5. Main User Scenarios

Scenario 1: user generates a new plan. Scenario 2: user submits feedback such as adding cardio or rest days. Scenario 3: user requests a goal-related nutrition/recovery tip. Scenario 4: admin/coach views users and their plans.

6. Main Routes

GET / — home page. POST /generate-workout — generate plan and tip. POST /submit-feedback — update a plan. GET /view-all-users — show stored user and plan records. GET /docs — interactive FastAPI API documentation.

7. Constraints and Assumptions

A valid Gemini API key and internet access are needed for live AI generation. The application is intended as general fitness guidance and does not replace advice from a qualified healthcare professional. Database records are stored locally using SQLite.

8. Acceptance Criteria

The home page loads; valid input generates a seven-day plan and tip; generated data is saved; feedback produces an updated plan; the admin view lists saved users/plans; invalid input and AI failures display a helpful message.

Note: This document describes the intended design and workflow. Test outcomes should be recorded only after the application has been run and verified.